

UNIVERSITY EXAMINATION

DATA STRUCTURES AND ALGORITHMS

Time Allowed: 1 Hour

Maximum Marks: 30

GENERAL INSTRUCTIONS:

- 1. Answer all questions according to section guidelines.
- 2. Figures to the right indicate full marks allocated to each question.

SECTION A - MULTIPLE CHOICE QUESTIONS

Select the correct alternative for each question.

- Q1.** Which of the following asymptotic notations represents the upper bound of an algorithm's running time? [3]
A. Omega B. Theta
C. Big-O D. Little-o
- Q2.** What is the worst-case time complexity of standard Binary Search on a sorted array of size n ? [3]
A. $O(n)$ B. $O(\log n)$
C. $O(n \log n)$ D. $O(1)$
- Q3.** Which of the following sorting algorithms has the best theoretical worst-case time complexity among comparison-based sorts? [3]
A. Bubble Sort B. Selection Sort
C. Insertion Sort D. Merge Sort
- Q4.** Which core property is essential for a problem to be successfully solved using a Greedy algorithm? [3]
A. Optimal substructure and greedy choice property B. Overlapping subproblems only
C. Random choice property D. Divide and conquer structure
- Q5.** Arrange the following growth rates in increasing order of magnitude: $O(n)$, $O(\log n)$, $O(n^2)$, $O(n \log n)$. [3]
A. $O(n)$, $O(\log n)$, $O(n \log n)$, $O(n^2)$ B. $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$
C. $O(\log n)$, $O(n \log n)$, $O(n)$, $O(n^2)$ D. $O(n^2)$, $O(n \log n)$, $O(n)$, $O(\log n)$
- Q6.** What is the primary reason QuickSort typically performs better in practice than MergeSort, despite having the same average time complexity? [3]
A. It is stable B. It requires no extra memory space (in-place sorting with good cache locality)
C. Its worst-case time complexity is $O(n)$ D. It does not use recursion

Q7. In the context of the Fractional Knapsack problem, why does the Greedy approach yield an optimal solution whereas it fails for the 0/1 Knapsack problem? [3]

- A. Items can be broken into smaller pieces, allowing the fractional value density to be fully exploited.
- B. Fractional knapsack has overlapping subproblems.
- C. 0/1 knapsack restricts item weights to powers of two.
- D. Greedy algorithms never fail for 0/1 knapsack.

Q8. Using Master Theorem, determine the time complexity of the recurrence relation $T(n) = 4T(n/2) + O(n^2)$. [3]

- A. $O(n^2)$
- B. $O(n^2 \log n)$
- C. $O(n^3)$
- D. $O(n \log n)$

Q9. Which classical algorithm solves the Single-Source Shortest Path problem using a greedy strategy and fails if edge weights are negative? [3]

- A. Bellman-Ford Algorithm
- B. Floyd-Warshall Algorithm
- C. Dijkstra's Algorithm
- D. Kruskal's Algorithm

Q10. What is the lower bound for comparison-based sorting algorithms in the worst case? [3]

- A. $O(n)$
- B. $O(n \log n)$
- C. $O(n^2)$
- D. $O(\log n)$

CONFIDENTIAL - OFFICIAL EVALUATION SCHEME & ANSWER KEY

SECTION A - Multiple Choice Questions - Solutions

Q1. (MCQ, 3M) [Topic: unit 1: Asymptotic notation | Level: Remember]

Answer/Rubric: C. Big-O - [Big-O notation is used to describe the asymptotic upper bound of a function, defining the worst-case growth rate.]

Q2. (MCQ, 3M) [Topic: unit 2: Searching and sorting | Level: Remember]

Answer/Rubric: B. $O(\log n)$ - [Binary search repeatedly divides the search interval in half, resulting in a logarithmic time complexity in the worst case.]

Q3. (MCQ, 3M) [Topic: unit 2: Searching and sorting | Level: Understand]

Answer/Rubric: D. Merge Sort - [Merge Sort guarantees a worst-case time complexity of $O(n \log n)$, whereas Bubble, Selection, and Insertion sorts take $O(n^2)$ in the worst case.]

Q4. (MCQ, 3M) [Topic: unit 3: Greedy algorithm | Level: Understand]

Answer/Rubric: A. Optimal substructure and greedy choice property - [Greedy algorithms require that a locally optimal choice leads to a globally optimal solution (greedy choice property) along with optimal substructure.]

Q5. (MCQ, 3M) [Topic: unit 1: Asymptotic notation | Level: Apply]

Answer/Rubric: B. $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$ - [Logarithmic grows slowest, followed by linear, linearithmic, and quadratic growth.]

Q6. (MCQ, 3M) [Topic: unit 2: Searching and sorting | Level: Analyze]

Answer/Rubric: B. It requires no extra memory space (in-place sorting with good cache locality) - [QuickSort sorts in-place and exhibits excellent cache reference locality, making it faster in practical implementations.]

Q7. (MCQ, 3M) [Topic: unit 3: Greedy algorithm | Level: Analyze]

Answer/Rubric: A. Items can be broken into smaller pieces, allowing the fractional value density to be fully exploited. - [Divisibility allows taking fractions of items to completely fill the knapsack capacity, satisfying the greedy choice property.]

Q8. (MCQ, 3M) [Topic: unit 1: Asymptotic notation | Level: Evaluate]

Answer/Rubric: B. $O(n^2 \log n)$ - [Here $a=4$, $b=2$, so $n^{(\log_b a)} = n^2$. Since $f(n) = O(n^2)$ and matches $n^2 \log^k n$ with $k=0$, Case 2 of the Master Theorem applies, yielding $O(n^2 \log n)$.]

Q9. (MCQ, 3M) [Topic: unit 3: Greedy algorithm | Level: Remember]

Answer/Rubric: C. Dijkstra's Algorithm - [Dijkstra's algorithm greedily selects the unvisited node with the smallest tentative distance, an assumption that breaks down in the presence of negative edge weights.]

Q10. (MCQ, 3M) [Topic: unit 2: Searching and sorting | Level: Evaluate]

Answer/Rubric: B. $O(n \log n)$ - [Based on the decision tree model, any comparison-based sorting algorithm must perform at least $\log(n!) = \Omega(n \log n)$ comparisons in the worst case.]